

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CHANDPUR SCIENCE AND TECHNOLOGY UNIVERSITY

Course Code: CSE 2201

Course Title: Algorithm Design and Analysis

Time: 30 Minutes

Full Marks: 20

Student ID:

Date:

CT-01

1. Which of the following best defines an algorithm?
 - A) A program written in a language
 - B) A flowchart
 - C) A step-by-step computational procedure
 - D) A compiler instruction
2. Pseudocode is:
 - A) A programming language
 - B) Machine code
 - C) High-level, language-agnostic algorithm representation
 - D) Only used in Java
3. Big-O notation represents:
 - A) Exact time
 - B) Best-case time complexity
 - C) Worst-case upper bound
 - D) Hardware-specific analysis
4. What is the time complexity of the following code?

```
for (i = 1; i <= n; i++) {  
    for (j = 1; j <= n; j++) {  
        sum += arr[i][j];  
    }  
}
```

 - A) $O(n)$
 - B) $O(\log n)$
 - C) $O(n^2)$
 - D) $O(1)$
5. Which of the following is the tight bound of $f(n) = 3n^2 + 7n + 8$?
 - A) $O(n)$
 - B) $\Theta(n^2)$
 - C) $\Omega(n^3)$
 - D) $O(\log n)$
6. What is the purpose of asymptotic analysis in algorithm design?
7. Define Big-O, Big- Ω , and Big- Θ notations in a single sentence each.
8. What is the practical significance of ignoring constant factors in time complexity analysis?
16. Determine the time complexity:

```
for (i = 0; i < n; i++) {  
    count++;  
}
```
17. Determine the time complexity:

```
for (i = 1; i <= n; i = i * 2) {  
    for (j = 1; j <= n; j++) {  
        sum += i * j;  
    }  
}
```
18. Show that $f(n) = 30n + 8$ is $O(n)$. Find suitable constants c and n_0 .